***Solution:***

| Tamil | | English | | $D_i = R_{1i} - R_{2i}$ | $D_i^2$ |
|---|---|---|---|---|---|
| Marks | Rank ($R_{1i}$) | Marks | Rank ($R_{2i}$) | | |
| 75 | 1 | 25 | 5 | -4 | 16 |
| 40 | 5 | 42 | 1 | 4 | 16 |
| 52 | 4 | 35 | 2 | 2 | 4 |
| 65 | 2 | 20 | 4 | -2 | 4 |
| 60 | 3 | 33 | 3 | 0 | 0 |
| | | | | | 40 |

$$\sum_{i=1}^{n} D_i^2 = 40 \text{ and } n = 5$$

$$\rho = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n\left(n^2 - 1\right)}$$

$$= 1 - \frac{6 \times 40}{5\left(5^2 - 1\right)} = 1 - \frac{240}{5(24)} = -1$$

**Interpretation:** This perfect negative rank correlation (-1) indicates that scorings in the subjects, totally disagree. Student who is best in Tamil is weakest in English subject and vice-versa.

## Example 4.5

Quotations of index numbers of equity share prices of a certain joint stock company and the prices of preference shares are given below.

| Years | 2013 | 2014 | 2015 | 2016 | 2017 | 2008 | 2009 |
|---|---|---|---|---|---|---|---|
| Equity shares | 97.5 | 99.4 | 98.6 | 96.2 | 95.1 | 98.4 | 97.1 |
| Reference shares | 75.1 | 75.9 | 77.1 | 78.2 | 79 | 74.6 | 76.2 |

Using the method of rank correlation determine the relationship between equity shares and preference shares prices.

***Solution:***

| Equity shares | Preference share | $R_{1i}$ | $R_{2i}$ | $D_i = R_{1i} - R_{2i}$ | $D_i^2$ |
|---|---|---|---|---|---|
| 97.5 | 75.1 | 4 | 6 | -2 | 4 |
| 99.4 | 75.9 | 1 | 5 | -4 | 16 |
| 98.6 | 77.1 | 2 | 3 | -1 | 1 |
| 96.2 | 78.2 | 6 | 2 | 4 | 16 |
| 95.1 | 79.0 | 7 | 1 | 6 | 36 |
| 98.4 | 74.6 | 3 | 7 | -4 | 16 |
| 97.1 | 76.2 | 5 | 4 | 1 | 1 |
| | | | | | $\sum_{i=1}^{n} D_i^2 = 90$ |



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